

ECON 1550

Spring 2026

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Submission: Canvas or Gradescope

Problem Set 3 Answer Key

1. Multiple Choice

Select only one answer.

(a) The carry trade

- ☐ (A) has positive average returns because the uncovered interest parity condition holds.
- ☐ (B) has a non-zero risk premium only when covered interest parity holds.
- ☐ (C) is risky because the difference between domestic and foreign interest rates fluctuates over time.
- ✓ **(D) earns a zero average risk premium when the uncovered interest parity condition holds.**

Solution:

- (A) Do not select. The carry trade has positive average returns because the uncovered interest parity condition does *not* hold empirically.
- (B) Do not select. The carry trade risk premium is about whether *uncovered* interest parity holds, not *covered* interest parity.
- (C) Do not select. The carry trade is risky because exchange rates fluctuate, not because interest rate differentials fluctuate.
- **(D) Select.** When UIP holds, expected returns from investing in foreign versus domestic bonds are equal, implying a zero average risk premium.

(b) Running a current account deficit is equivalent to

- ✓ **(A) net borrowing from the rest of the world.**
- ☐ (B) net lending to the rest of the world.

- ☐ (C) an increase in public savings.
- ☐ (D) an increase in private savings.

Solution:

- **(A) Select.** A current account deficit means imports exceed exports, so the country must finance the difference by borrowing from abroad (or selling assets to foreigners).
- (B) Do not select. Lending to the rest of the world corresponds to a current account *surplus*, not a deficit.
- (C) Do not select. An increase in public savings (larger government budget surplus or smaller deficit) would tend to improve the current account, not cause a deficit.
- (D) Do not select. An increase in private savings would also tend to improve the current account, not cause a deficit.

(c) If the government has a \$100 million budget deficit, private saving is equal to \$500 million, private investment is equal to \$300 million, what is the value of the current account?

- ✓ **(A) \$100 million surplus.**
- ☐ (B) \$700 million surplus.
- ☐ (C) \$100 million deficit.
- ☐ (D) \$700 million deficit.

Solution:

- **(A) Select.** Using the identity $S^p + (T - G) = I + CA$, we have $500 + (-100) = 300 + CA$, so $CA = 100$ million surplus.
- (B) Do not select. This would require a different combination of savings, investment, and government balance.
- (C) Do not select. This has the wrong sign; the calculation yields a surplus, not a deficit.
- (D) Do not select. This has both the wrong sign and the wrong magnitude.

(d) Which of the following correctly shows the relationship between savings, the government budget balance, and the current account?

- ☐ (A) $S^p + CA = I + (T - G)$
- ☐ (B) $S^p + CA = I + (T + G)$
- ✓ ☒ (C) $S^p + (T - G) = I + CA$
- ☐ (D) $S^p + (T + G) = I + CA$

Solution:

- (A) Do not select. The terms are on the wrong sides of the equation.
- (B) Do not select. Both the placement and the sign of G are incorrect.
- **(C) Select.** This is the correct national income identity: private savings plus public savings (government budget balance $T - G$) equals investment plus the current account.
- (D) Do not select. The government budget balance should be $T - G$, not $T + G$.

(e) All else equal, if Canada raises its interest rates,

- ✓ ☒ (A) **the U.S. dollar depreciates.**
- ☐ (B) the U.S. demand for Canadian dollars decreases.
- ☐ (C) the Canadian supply of Canadian dollars increases.
- ☐ (D) the Canadian dollar will depreciate.

Solution:

- **(A) Select.** Higher Canadian interest rates make Canadian assets more attractive, increasing demand for CAD. This causes the CAD to appreciate and the USD to depreciate.
- (B) Do not select. U.S. demand for Canadian dollars *increases* as U.S. investors seek higher Canadian interest rates.
- (C) Do not select. Canadians have less incentive to convert CAD to foreign currencies when domestic rates rise, so the supply of CAD decreases.
- (D) Do not select. Higher interest rates attract capital inflows, causing the

Canadian dollar to *appreciate*, not depreciate.

2. Chapter 3: Exchange Rates and the Foreign Exchange Market: An Asset Approach

- (a) Please answer question 11 from Chapter 3 of the textbook, reproduced here for convenience:

“Suppose the dollar exchange rates of the euro and the yen are equally variable. The euro, however, tends to depreciate unexpectedly against the dollar when the return on the rest of your wealth is unexpectedly high, while the yen tends to appreciate unexpectedly in the same circumstances. As a U.S. resident, which currency, the euro or the yen, would you consider riskier?”

Solution: The euro is less risky for you. When the rest of your wealth falls, the euro tends to appreciate, reducing your losses by giving you a relatively high payoff in terms of dollars. Losses on your euro assets, on the other hand, tend to occur when they are least painful, that is, when the rest of your wealth is unexpectedly high. Holding the euro, therefore, reduces the variability of your total wealth.

This logic showed up as one of the possible explanations for why the carry trade has a non-zero risk premium.

- (b) Consider our model of exchange rate determination through uncovered interest parity (UIP) for two time periods, t and $t + 1$. Tables I and II summarize this two-period model. The behavioral equation for the expected exchange rate at $t + 1$, E_{t+1}^e , is missing from Table II (the entry for that equation is ‘??’). Propose a reasonable behavioral equation for E_{t+1}^e such that an (exogenous) increase in E_t^e results in an increase in E_{t+1}^e . Explain the intuition for why E_{t+1}^e increases when E_t^e increases.

Hint: By reasonable, we mean reasonable to you. This is your own new model of exchange rate determination!

Table I: Exogenous variables	
Variable	Description
R_t	Domestic interest rate at t
R_t^*	Foreign interest rate at t
E_t^e	Expected exchange rate at t
R_{t+1}	Domestic interest rate at $t + 1$
R_{t+1}^*	Foreign interest rate at $t + 1$

Table II: Endogenous Variables and Equations			
Variable	Description	Equation	Type of equation
E_t	Exchange rate at t	$R_t = R_t^* + \frac{E_t^e - E_t}{E_t}$	Equilibrium condition
E_{t+1}	Exchange rate at $t + 1$	$R_{t+1} = R_{t+1}^* + \frac{E_{t+1}^e - E_{t+1}}{E_{t+1}}$	Equilibrium condition
E_{t+1}^e	Expected exchange rate at $t + 1$	??	Behavioral equation

Solution: We propose one particular answer, but of course there are many correct answers.

Irrespective of what equation we propose for E_{t+1}^e , the equilibrium at t looks like the standard equilibrium of the static (one-period) model we studied in class. For this standard model, we can solve the UIP condition for E_t to get

$$E_t = \frac{E_t^e}{1 + R_t - R_t^*}$$

When E_t^e goes up, E_t goes up as well: at time t , the expectation of a future depreciation leads to an actual depreciation.

For E_{t+1} to also increase in response to the increase in E_t^e , we have to make E_{t+1}^e depend on E_t^e , E_t , or both. Let's try a behavioral equation for E_{t+1}^e given by:

$$E_{t+1}^e = E_t^e$$

One way to justify this equation with economic intuition is to interpret the expected exchange rate for both t and $t + 1$ as expectations for the value that the exchange rate will take “in the long run”, many, many periods after t and $t + 1$.

To the extent that the long run looks the same from the perspectives of times t and $t + 1$ (since one period is not a big difference when thinking about the far future), revisions in expectations about the long run at t should lead to an equal revision at $t + 1$. The idea is that we will not change our mind about the long run in just one period.

$$E_{t+1} = \frac{E_{t+1}^e}{1 + R_{t+1} - R_{t+1}^*} = \frac{E_t^e}{1 + R_{t+1} - R_{t+1}^*}$$

The expectation of a long-run depreciation at time t (E_t^e goes up) produces an equal expectation of a long run depreciation at $t + 1$ (E_{t+1}^e goes up due to our behavioral equation $E_{t+1}^e = E_t^e$).

From here, the intuition is the one of the standard model.

Keeping the exchange rate E_{t+1} fixed at its initial value (before the change in E_t^e), an expected depreciation increases the domestic-currency return of investing in the foreign bond. After converting domestic currency into foreign currency at the given exchange rate E_{t+1} and earning the foreign interest rate R_{t+1}^* , the higher expected exchange rate E_{t+1}^e implies that the same amount of foreign currency earned by investing in the foreign bond is now exchanged into a larger amount of domestic currency.

The domestic-currency return of investing in the domestic bond, R_{t+1} , has not changed.

We have found that, at the initial exchange rate, the domestic-currency return on the foreign bond is higher than that of the domestic bond. But equilibrium requires that the two are equal.

To reduce the domestic-currency return of investing in the foreign bond, the exchange rate must depreciate today (E_{t+1} has to go up). When E_{t+1} goes up, the same amount of domestic currency results in a smaller amount of foreign currency that can be invested in the foreign bond. Even though the foreign-currency return on the foreign bond, R_{t+1}^* , has not changed, the smaller initial investment leads to a smaller payoff. When the reduction in payoff due to the higher exchange rate offsets the increase in payoff due to the higher expected exchange rate, equilibrium is reached.

(c) Question 18 from Chapter 3 of the textbook states:

“The interest rate on U.S. three-month Treasury bills dropped to very low

levels at the end of 2008 and remained there for several years. Starting in January 2009 and ending in December 2019, find data on the three-month Treasury bill rate from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis; find data on the exchange rate of the U.S. dollar against the Korean won from the Bank of Korea Economic Statistics System at http://ecos.bok.or.kr/flex/EasySearch_e.jsp; and from the same source, find data on the Korean 91-day Monetary Stabilization Bond interest rate. Imagine that you borrow dollars at the Treasury bill rate to invest in Korean stabilization bonds, thus doing a carry trade that exposes you to the risk of won/dollar exchange rate fluctuations. As in the Case Study in the text, calculate the total return on your carry trade for every month starting in February 2009 and ending in December 2019.”

Please answer this question but end your analysis in July 2023 (rather than on the earlier date in the textbook’s question). In addition to calculating the returns, please include a plot that shows the calculated total returns¹ (on the vertical axis) for each month (on the horizontal axis) assuming your initial investment was \$100.

Note also that the link http://ecos.bok.or.kr/flex/EasySearch_e.jsp to find data from the Bank of Korea Economic Statistics System provided by the textbook is outdated and no longer works. The correct link is:

<https://ecos.bok.or.kr/#/SearchStat>

Be patient on the first load, it takes a while to download the page but works well after that.

Hint: You can review how to compute payoffs and returns for the carry trade in [these lecture slides](#).

Solution: There are two steps to this question. The first step is getting the data, the second step is constructing the carry trade returns.

Getting the data

- Bank of Korea

To get the Monetary Stabilization Bond (MSB) interest rate, go to:

<https://ecos.bok.or.kr/#/SearchStat>

¹Total returns are the same as cumulative returns. For example, if your initial investment is \$100 in January of 2023 and the value of your portfolio is \$120 in March 2023, your total or cumulative return is $120/100 - 1 = 0.2 = 20\%$. This 20% return can be achieved with different combinations of monthly (not cumulative) returns for the months of January, February, and March.

Then, in the Table Select pane, navigate to:

1. Monetary Financial Statistics > 1.3. Interest Rates > 1.3.2. Market Interest Rates > 1.3.2.2. Market Interest Rates (Monthly, Quarterly, Annual)

Once there, the Item Select pane will be populated with a list of rates. Turn off the “Select All” switch at the top of the pane, then from the list select “Monetary stabilization bonds (91-day)”. At the bottom of the panel, click the “Add to List” button. Now the selected series appears on the “Table Name” pane.

To get the exchange rate, there are two equally correct options.

- Option 1. In the Table Select pane, navigate to:
 3. Exchange Rate/International Reserves and Trade > 3.1. Foreign Exchange Rate > 3.1.2 Average Period, End Period > 3.1.2.1. Arbitraged Rates of Major Currencies Against Won, Longer Frequency
- Option 2. In the Table Select pane, navigate to:
 3. Exchange Rate/International Reserves and Trade > 3.1. Foreign Exchange Rate > 3.1.2 Average Period, End Period > 3.1.2.3. Exchange Rate of Won Against US Dollar, China Yuan Renminbi [...]

The exchange rates from the two options are very close to each other². Then follow the same steps as before to add “Won per United States Dollar (Close)” to the Table Name pane. You can use either the “Closing Rate” or the “Average Rate” for the “Won per United States Dollar (Close)” series. They are both equally good options and very close to each other. In this answer, we use the “Closing Rate” from option 1 above.

Last, at the bottom of the Table Name pane, click “View List”. A new screen will appear. Check the box for “Mon” (for monthly series) and select 2009.01 as the starting date and 2023.07 as the ending date. After selecting the dates, you have to click “search”, otherwise the dates will not be updated. Finally, customize the format if desired (e.g., vertical view rather than horizontal) and click on “Original Data Download”.

- FRED

The three-month Treasury bill rate from FRED has code TB3MS. However, the series “Market Yield on U.S. Treasury Securities at 3-Month Constant Maturity, Quoted on an Investment Basis” that has code GS3M is equally appropriate. In this answer, we use TB3MS.

The original spreadsheets with the downloaded data from the Bank of Korea can be found [here](#) (or in [PDF](#)) and from FRED it can be found [here](#) (or in [PDF](#)).

Constructing carry trade returns

- Formula for returns

The carry trade returns are the same whether we compute US Dollar (USD) returns or Korean Won (KRW) returns. We compute USD returns.

We use the following notation:

E_t = USD / KRW spot exchange rate (Dollars per Won) in month t

R_t = Interest rate for the USD-denominated US Treasury bill in month t

R_t^* = Interest rate for the KRW-denominated Korean bond in month t

For the time- t interest rates R_t and R_t^* , we note that the returns accrue between times t and $t + 1$ (this notation follows the convention that variables are known at the time of their subscript, but not earlier).

The payoff in USD earned by investing 1 USD in the KRW bond (between t and $t + 1$) are:

$$(1 + R_t^*) \times \frac{E_{t+1}}{E_t} - 1$$

The USD returns earned by investing in the KRW bond (between t and $t + 1$) are:

$$(1 + R_t^*) \times \frac{E_{t+1}}{E_t} - 1$$

The USD returns earned by borrowing in the USD bond (between t and $t + 1$) are:

$$-R_t$$

The negative sign is there because we are borrowing (short-selling the bond) rather than lending (buying the bond).

The returns on the carry trade strategy are therefore:

$$\begin{aligned}\text{returns}_t &= \left[(1 + R_t^*) \times \frac{E_{t+1}}{E_t} - 1 \right] - R_t \\ &= (1 + R_t^*) \times \frac{E_{t+1}}{E_t} - (1 + R_t)\end{aligned}$$

We can also use that the USD returns earned by investing in KRW bonds are approximately³:

$$R_t^* + \left(\frac{E_{t+1}}{E_t} - 1 \right)$$

Which gives

$$\begin{aligned}\text{Approximate returns}_t &= R_t^* + \left(\frac{E_{t+1}}{E_t} - 1 \right) - R_t \\ &= R_t^* - R_t + \left(\frac{E_{t+1}}{E_t} - 1 \right)\end{aligned}$$

The term $R_t^* - R_t$ is the *interest rate differential* and the term $E_{t+1}/E_t - 1$ is the *realized depreciation rate* of the USD with respect to the KRW.

Both the exact and the approximate returns can now be computed using the data on R_t^* and E_t downloaded from the Bank of Korea and the data on R_t downloaded from FRED.

- Formula for cumulative returns

To compute cumulative returns, we take January 2009 to be month $t = 0$ and set the initial value of cumulative returns to be 100:

$$\text{Cumulative returns}_0 = 100$$

For $t > 0$, we compute cumulative returns using the recursive formula:

$$\text{Cumulative returns}_t = \text{Cumulative returns}_{t-1} \times (1 + \text{returns}_t)$$

- Converting to the Correct Units

Our data sources provided us with returns in units of *annualized percentage points*. However, R and R^* in our calculations are *monthly net returns* (net here just means not in percentage points, so a 1% return corresponds to a net re-

turn of 0.01). To convert the annualized percentage point returns that the data sources provided into the net monthly returns that we use in our formulas above, we have to divide by 12 (because there are 12 months in a year) and by 100 (to go from percentage points to net). For example, if the original return we downloaded is 2.7, the monthly net return is $2.7/12/100 = 0.00225$.

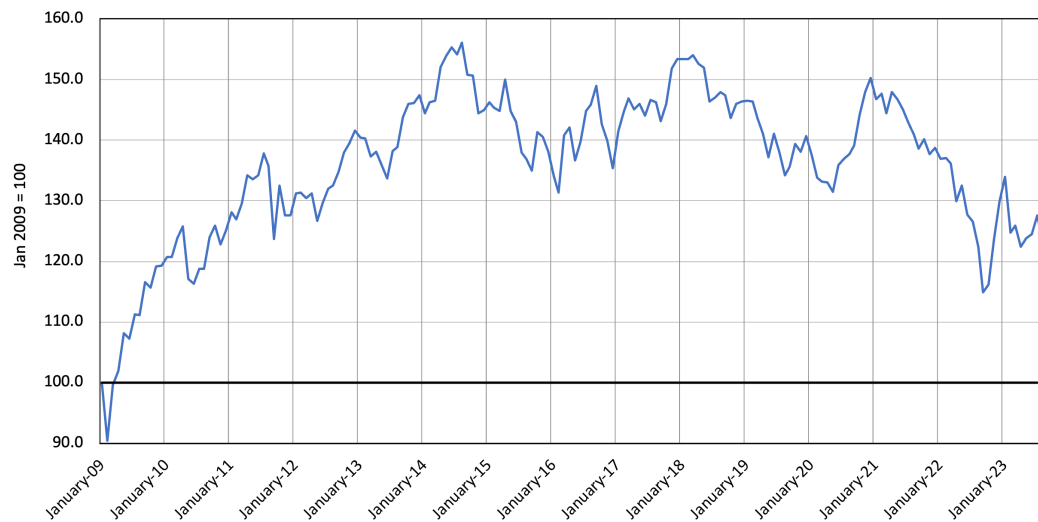
In addition, our formulas have the convention that R_t and R_t^* are returns accrued between t and $t + 1$. However, the data sources use the convention that R_t and R_t^* are returns accrued between $t - 1$ and t . Therefore, before we can use the returns provided by our data sources, we have to shift them by one month. For example, to compute carry trade returns between January and February, we have to use interest rates that our data sources provide as February interest rates.

For the exchange rate, our data sources give us the KRW / USD (Wons per Dollar) exchange rate. However, in our calculations, we use E_t , which is the USD / KRW exchange rate (Dollars per Won). Therefore, to convert the exchange rate we downloaded into E_t , we have to take its reciprocal. For example, if we downloaded an exchange rate value of 1,250 Won per Dollar, then

$$E_t = \frac{1}{1,250 \text{ KRW/USD}} = \frac{1}{1,250} \text{ USD/KRW} = 0.0008 \text{ Dollars per Won}$$

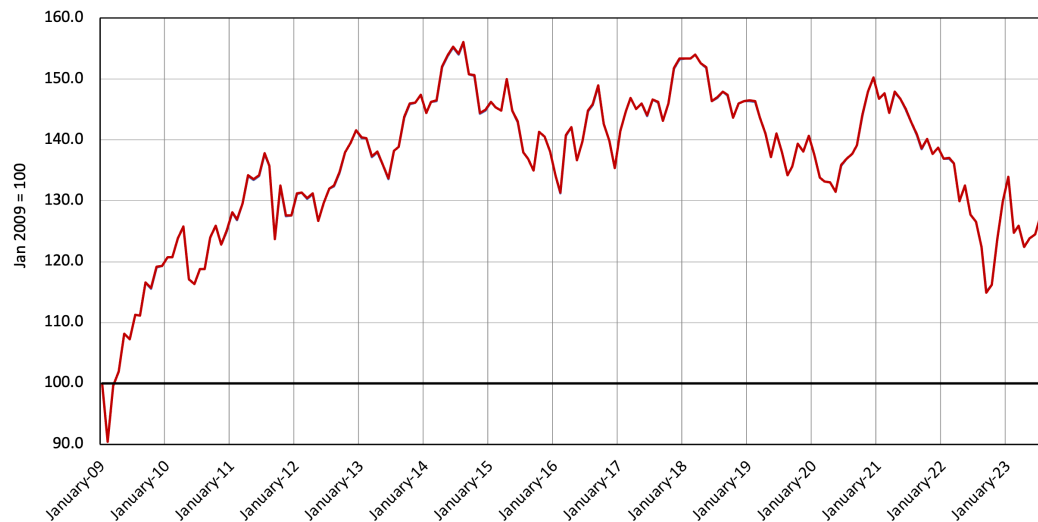
The plot below shows the time series of cumulative returns. The initial value of 100 that we picked is arbitrary – we could have used any initial value (except zero). You can think of cumulative returns in month t as the amount of money you would have in month t if you had invested \$100 in the carry trade strategy in January 2009.

Cumulative return on carry



If we use approximate returns rather than returns, we get a very similar plot:

Approximate cumulative return on carry



You can see all the calculations and plots in [this Excel file](#).

²The exchange rate in Option 1 is the “basic exchange rate” and is determined as the transactions volume-weighted average of the rates applied in the previous business day’s transactions between foreign exchange banks through brokers in an “over-the-counter” (OTC) market. The exchange rate in Option 2 is the closing-day exchange rate quoted in a trading exchange (a marketplace where prices are quoted openly).

³Start with exact returns

$$\begin{aligned}(1 + R_t^*) \times \frac{E_{t+1}}{E_t} - 1 &= (1 + R_t^*) \times \left(\frac{E_{t+1}}{E_t} - 1 + 1 \right) - 1 \\&= (1 + R_t^*) \times \left[\left(\frac{E_{t+1}}{E_t} - 1 \right) + 1 \right] - 1 \\&= (1 + R_t^*) \times \left(\frac{E_{t+1}}{E_t} - 1 \right) + (1 + R_t^*) - 1 \\&= (1 + R_t^*) \times \left(\frac{E_{t+1}}{E_t} - 1 \right) + R_t^* \\&= \left(\frac{E_{t+1}}{E_t} - 1 \right) + R_t^* \times \left(\frac{E_{t+1}}{E_t} - 1 \right) + R_t^*\end{aligned}$$

Using that $R_t^* \times \left(\frac{E_{t+1}}{E_t} - 1 \right) \approx 0$ is a good approximation when both R_t^* and $\left(\frac{E_{t+1}}{E_t} - 1 \right)$ are close to zero, we get:

$$\left(\frac{E_{t+1}}{E_t} - 1 \right) + R_t^* \times \left(\frac{E_{t+1}}{E_t} - 1 \right) + R_t^* \approx \left(\frac{E_{t+1}}{E_t} - 1 \right) + R_t^*$$